WESTMINSTER
International University in Tashkent

System Analysis & Design

Lecture 6

Agenda

WIUT

1. Last week recap
    1. Extreme Programming (XP)
2. Spiral Model
3. Questions
4. Recommended reading

WESTMINSTER
International University in Tashkent

An Accredited Institution of
the University of Westminster (UK)

Recap

What did we discuss last week?

XP: Values

WIUT

1. **Communication** – promote among the team members, also involve end users actively.
2. **Simplicity** – design the simplest solution. Code for today, not tomorrow. "You ain’t gonna need it" (YAGNI) approach.
3. **Feedback** – from unit tests, acceptance tests, customer.
4. **Courage** – to change your code or throw it away.
5. **Respect** – your peers. Don’t commit a change that breaks tests or compilation. Make sure everyone feels appreciated and heard.

XP: Practices

WIUT

Fine-scale feedback

- Pair programming
- Planning game
- Test-driven development
- Whole team

Continuous process

- Continuous integration
- Refactoring or design improvement
- Small releases

Shared understanding

- Coding standards
- Collective code ownership
- Simple design
- System metaphor

Programmer welfare

- Sustainable pace

XP: Rules

WIUT

Planning

- User stories are written.
- Release planning creates the release schedule.
- Make frequent small releases.
- The project is divided into iterations.
- Iteration planning starts each iteration.

Managing

- Give the team a dedicated open stand-up.
- Set a sustainable pace.
- A stand-up meeting starts each day.
- The Project Velocity is measured.
- Move people around.
- Fix XP when it breaks.

Design

- Simplicity.
- Choose a system metaphor.
- Use CRC cards for design sessions.
- Create spike solutions to reduce risk.
- No functionality is added early.
- Refactor whenever and wherever possible.

Coding

- The customer is always available.
- Code must be written to agreed standards.
- Code the unit test first.
- All production code is pair programmed.
- Only one pair integrates code at a time.
- Integrate often.
- Set up a dedicated integration computer.
- Use collective ownership.

Testing

- All code must have unit tests.
- All code must pass all unit tests before it can be released.
- When a bug is found tests are created.
- Acceptance tests are run often and the score is published.

WIUT

Before we move on,
let me tell you a story…

Lessons learned…

WIUT

1. **Projects fail and fail a lot** – Remember Research paper from Tutorial 1.

2. **There are many factors that lead to project failure:**
    1. Not clear, vague or changing requirements
    2. Undefined project scope
    3. Financial, technical, human resource aspects
    4. Developers postpone
    5. etc.

3. **What should we do?** – Manage our projects and the whole process.

4. **How can we do it?** – Use development methodologies that fit our context and address problems that we have (i.e. use them for their “side effects”).

Biggest problems…

WIUT

1. Not clear, vague or changing requirements is one of the biggest problems in real life software development.

2. How can we address this problem?
   What can we do to reduce the risk of project failure that is caused by **(a)** vague or **(b)** changing requirements?
   a) Devote a stage for requirements gathering; use different techniques we covered to gather quality requirements (involve end-users, prototyping, etc.); document requirements;
   b) Split the project into smaller parts; start with the most important functionalities; write system that considers possible code changes; etc.

New/Modern methodologies

WIUT

- So, this is where many of the Incremental and Iterative models, as well as many of the new/modern methodologies come in.
- By splitting projects into smaller pieces, it is easier to manage and change them (compared to a big plan ahead).
- All methodologies, that we will see, base their ideas on this simple premise: divide and concur.
- Every project has a risk of failure. We need to manage risk. Spiral model puts emphasis on evaluating and addressing risk.

WESTMINSTER
International University in Tashkent

An Accredited Institution of
the University of Westminster (UK)

Spiral Model

One of the first iterative models.

System Develop Methodologies

WIUT

Structural:

- Waterfall
- Structured Software Development Methodology (SSDM)

Iterative and Incremental:

- Spiral
- RAD
- eXtreme Programming
- Scrum
- Etc.

Prototyping:

- Prototyping

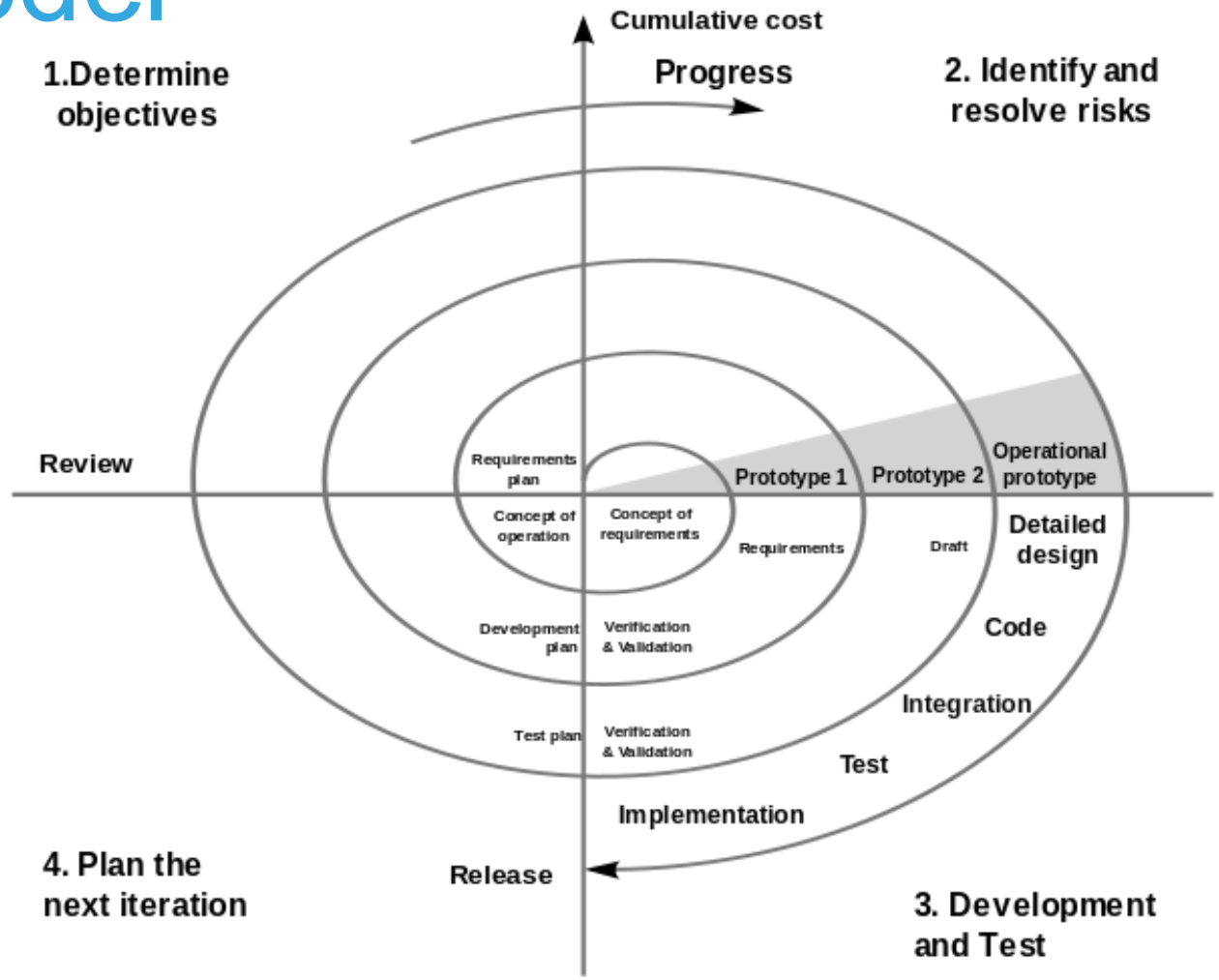
Spiral Model

WIUT

- Developed in 1986 by Barry Boehm.
- Puts emphasis into addressing risks.
- Applies iterative approach.
- You can think of it as Prototyping + Waterfall.

Spiral Model

WIUT



Source: "Spiral model (Boehm, 1988)" Licensed under Public Domain via Commons - https://commons.wikimedia.org/wiki/File:Spiral_model_(Boehm,_1988).svg

Spiral Model

WIUT

- Divide project into smaller ones.

- Iterate through each using the following stages:
  1. Identify objectives, alternatives, and constraints;
  2. Evaluate alternatives, identify and resolve risks;
  3. Develop and verify deliverables;
  4. Plan the next iteration.

Discussion

WIUT

- What are the advantages of adopting the methodology?
- What are the disadvantages of using the methodology?



WESTMINSTER
International University in Tashkent

An Accredited Institution of
the University of Westminster (UK)

Questions?

Coursework 1 Q&A.

Recommended Reading

WIUT

- Wikipedia, (2015). *Spiral model*. [online] Available at: https://en.wikipedia.org/wiki/Spiral_model [Accessed 26 Oct. 2015].

- *Google…*



The End